MIL-C-10856D
20 September 1966
SUPERSEDING
MIL-M-10856C
29 December 1961

MILITARY SPECIFICATION

CHEMICAL AGENT, MUSTARD, SIMULATED

(MOLASSES RESIDUUM)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of simulated mustard (molasses residuum).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-E-485 - Enamel, Semi-Gloss, Rust-Inhibiting.
PPP-D-729 - Drums; Metal, 55-Gallon (for Shipment of Noncorrosive Material).
PPP-P-420 - Plugs and Flanges (For Drum Closure).

MILITARY

MIL-V-12276 - Varnish, Phenolic, Baking.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-171 - Finishing of Metal and Wood Surfaces.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these freight classification ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1193-56 - Reagent Water.

(Application for copies of the above publication should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The molasses residuum shall be a concentrate of stillage from fermentation of molasses treated to prevent further fermentation. The material shall not produce gas when tested as specified in 4.5.1 (see 6.4).

3.2 Chemical and physical requirements. The molasses residuum shall conform to the physical and chemical requirements of table I.



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Table I. Chemical and physical requirements

| Property | Requirement | Test method |
|---|---|---|
| Density at 15.6°C (min) | 1.20 | 4.5.2 |
| Evaporation loss, percent (max) | 60 | 4.5.3 |
| Ash content, percent (max) | 25 | 4.5.4 |
| Acidity calculated as sulfur dioxide, percent (max) | 3.5 | 4.5.5 |

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 and for other requirements for which specific inspection procedures have not been given are satisfied.

4.2 Lotting. A lot shall consist of the molasses residuum offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot. (See 6.3.)

4.3 Sampling.

4.3.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.



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4.3.2 For test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2. An approximately equal portion of the molasses residuum shall be taken from each container in the sample, composited, and placed in a clean, dry container labeled to identify the lot and the container from which it was taken. The total composited sample shall be 1/2 to 3/4 pounds.

4.4 Examination. Sample container of molasses residuum shall be examined for compliance with level A preparation for delivery requirements in accordance with the classification of defects and with MIL-STD-105.

4.4.1 For test. The composited specimen of molasses residuum shall be tested in accordance with 4.5. If the composited specimen fails any test specified therein, the lot represented shall be rejected.

4.4.2 Classification of defects.

(a) Preparation for delivery (section 5).

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | AQL 1.5 percent defective |
| 101 | Container not as specified |
| 102 | Container leaking or damaged |
| 103 | Marking missing, illegible, or incorrect |
| 104 | Interior coating missing or inadequate |

4.5 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.5.1 Fermentation. Dilute one part of the composited sample with three parts of sterile water and add 1 gram (g) of compressed yeast to 50 milliliters (ml) of the dilute mixture. Mix the suspension thoroughly and place 15 ml in a sterile Smith (or similar) fermentation tube. In filling the tube, care should be exercised to avoid the presence of air bubbles in the closed arm. Incubate the tube for 24 hours at 30° ± 2°C and examine visually for the presence of gas in the upper end of the closed arm.



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4.5.2 Density. Determine the density of 15.6° ± 1°C by means of a Westphal balance or hydrometer.

4.5.3 Evaporation loss. Tare to the nearest milligram (mg) a flat-bottom dish containing a short glass stirring rod and 6 to 7 g of pure quartz sand (previously digested with concentrated hydrochloric acid, washed, dried, ignited, cooled, and stored in a desiccator). Weigh approximately 3.5000 g of specimen into the tared dish, stir into the sand, and dry at 105° ± 5°C for 8 to 10 hours. Stir, reheat at 105° ± 5°C for 1 hour, cool, and reweigh. Repeat heating and weighing until loss in weight in 1 hour is no greater than 3 mg. Calculate the percent evaporation loss as follows:

$$\text{Percent evaporation loss} = \frac{100\ (A + B - C)}{B}$$

where: A = weight of dish, rod, and sand, in grams,
B = weight of specimen, in grams, and
C = weight of dish, rod, sand, and residue, in grams.

4.5.4 Ash. Weigh, to the nearest mg, approximately 5 g of the sample and heat in a 50- or 100-ml platinum dish until all water is expelled. Add a few drops of pure olive oil and heat slowly over a flame until swelling ceases. Place the dish in a muffle furnace and heat at approximately 500°C until a white ash is obtained. Treat the residue with approximately 5 ml of a 10-percent ammonium carbonate solution. Reevaporate and heat in the muffle furnace at approximately 450°C to constant weight. Calculate the percent ash as follows:

$$\text{Percent ash} = \frac{100\ A}{B}$$

where: A = grams of residue, and
B = grams of specimen.

4.5.5 Acidity. Weigh, to the nearest mg, approximately 4 g of the sample into a suitable container. Dilute to about 200 ml with water and determine the acidity by titrating electrometrically with 0.1N sodium hydroxide (NaOH). Calculate the acidity as sulfur dioxide ($SO_2$) as follows:

$$\text{Percent acidity as } SO_2 = \frac{3.203\ VN}{W}$$

where: N = normality of NaOH solution,
V = milliliters of NaOH solution, and
W = grams of specimen.



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## 5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Fifty-three gallons ± one gallon of molasses residuum shall be packed in a 55-gallon marked capacity steel drum conforming to type IV of PPP-D-729. Each removable head shall be provided with one 2-inch and one 3/4-inch flange with gasketed plug, diametrically opposed. The flanges and plugs shall conform to type I of PPP-P-420. All inner and outer surfaces of the drum including those of the locking ring and associated hardware shall be zinc phosphate coated in accordance with the requirements for phosphate coating of PPP-D-729. In addition, the inner surface of the drum including that of the removable head, shall be coated with a phenolic baking varnish film, of a minimum thickness of 2 mils, conforming to type II of MIL-V-12276; and the outer surfaces of the drum, including those of the removable head, the locking ring and associated hardware shall be coated with enamel conforming to type I, II, or III of TT-E-485, nitrogeneous resin modified. Interior coating shall be applied in accordance with the general painting requirements for organic finishes of MIL-STD-171. Exterior coating shall be applied in accordance with requirements for finish No. 21.5 for semi-gloss paint finishes for metal surfaces of MIL-STD-171.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with MIL-STD-129. Containers should bear a caution label to show that this material contains cresol and should not be swallowed.

## 6. NOTES

6.1 Intended use. The molasses residuum covered by this specification is intended for use in training as a simulant for mustard gas (HD).

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or some physical mixing operation intended to make the final product substantially uniform.



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6.4 Fermentation. It has been found that molasses residuum containing 1.6 percent cresol will meet the nonfermentation requirement when tested in accordance with 4.5.1.

Custodians:

Army - MU
Navy - OS

Review interest:

Army - MD
Navy - OS

User interest:

Navy - AS, MC

Preparing activity:

Army - MU (EA)

Project No. 1365-0140



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

**DD FORM 1 JAN 66 1426** REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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